

Fig. 4: Beacon network

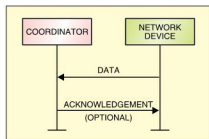


Fig. 5: Non-beacon network

in around 15 milliseconds, whereas Bluetooth in the same state takes three seconds to wake up and respond.

The network range for Bluetooth is 10 metres, whereas ZigBee has a range of 75-100 metres. Bluetooth is limited to eight nodes per network, while ZigBee can support up to 255 nodes.

Bluetooth battery requires frequent recharging, whereas ZigBee battery can last for years. Although ZigBee offers less bandwidth than Bluetooth, it consumes far less power, making it suitable for devices that need to constantly transfer data.

Applications

ZigBee networks could be used in homes, buildings and industries for automation of lighting control, heating, ventilation, air-conditioning, security, etc (see box on next page). Basically, ZigBee is the only standard-based technology that addresses the unique needs of most remote monitoring and control network applications.

ZigBee-ready devices

Mobile phones with ZigBee solution. Pantech&Curitel launched the world's

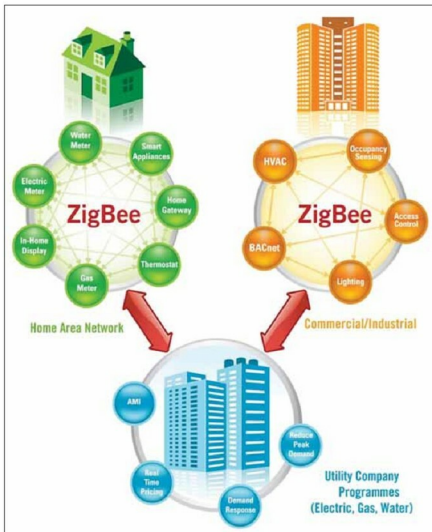


Fig. 6: ZigBee applications

first ZigBee phone at *ZigBee Alliance Festival*. The ZigBee-applied P1 of Pantech&Curitel provides intellectual home/building networking services such as controlling electrical appliances, alarming mail receipt, checking temperature and humidity and sending mobile messages to alarm in case of trespass. Moreover, with location-based services, the ZigBee phone can give more precise information on the route than the existing GPS function. For instance, it provides the names of restaurants and even the menu or price list of the chosen restaurant without requiring wireless Internet connection.

ZigBee wireless thermostats. Airbee has announced the first ZigBee

thermostat which includes an intelligent software for better home monitoring. It saves customers' money by not wasting resources that are not needed. Installers and contractors will benefit from easier installation, because by using ZigBee there would be no wires to connect up.

ZigBee-compatible general devices. ZigBee devices demonstrated by a company in San Francisco include a key hanger that helps to keep track of keys. The key hanger wirelessly connects to the user's mobile phone and when the two go out of range, the mobile phone warns its owner that he has lost his keys. The phone is able to tell whether it is at home, office or some outside location by connecting to other